

MUHAMMAD TOQEEER

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Summary

Proactive Data Scientist with a strong background in NLP, Computer Vision, and Large Language Models (LLMs). I specialize in end-to-end pipeline development, model training, and innovative problem-solving. My passion lies in harnessing cutting-edge technology, including LLMs, to spearhead data-driven innovation within the tech industry.

Experience

Addo AI

DATA SCIENTIST

Lahore, Pakistan

March 2024 - present

- Deployed a speech-to-text model for real-time transcription on SageMaker reduced response time 1.5 sec for 20 sec audio
- Conducted replenishment planning for apparel through forecasting and optimization.
- Optimized transportation from pit-to-port and the material management chain.

chainGPT

MACHINE LEARNING ENGINEER

US

March 2023 - March 2024

- Developed an LLM for code generation that generates smart contracts.
- Developed a general-purpose chatbot for answering questions in the crypto world.
- Created a RAG-based chatbot that communicates with live APIs to provide real-time information.
- Created RESTful APIs using Flask/FAST to enhance user experience and accessibility.

Funavary Technologies

SOFTWARE ENGINEER DATA SCIENTIST

Islamabad, Pakistan

Dec 2022 - March 2024

- End-to-end Architecture design and development of product and deployment
- Collaborate with product owners to ensure project goals are met
- Attend weekly and bi-weekly meetings with product owners
- Analyze test results and identify areas for improvement in the application and testing processes

Education

Masters of Science in Data Science

NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES, ISLAMABAD, PAKISTAN

2021- 2023

Bachelors of Science in Computer Science

NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES, ISLAMABAD, PAKISTAN

2016 - 2020

Projects

AI Smart Contracts (chainGPT.org)

2024

- Developed a Large Language Model (LLM) that generates smart contracts based on user-defined prompts.
- Created an end-to-end pipeline for LLM training, focusing on the initial stages of scraping smart contracts and parsing them to generate training prompts.
- Performed distributed training of the LLAMA-2 model with full parameters.
- Deployed models on AWS SageMaker for scalable use.

Framework: Pytorch, LLM, LLAMA-2, ChatGPT, Transformers, FastAPI

- Developed a domain-specific chatbot for navigating the crypto and blockchain landscape.
- Built on a dataset covering crypto, blockchain, and Web3 industries.
- Utilized RAG technique for context-aware responses by fetching relevant data from a vector database.
- Equipped the AI to process and evolve with historical and current industry knowledge.

Framework: PyTorch, Transformers, LLMs, Blockchain, OpenAI, Langchain, FASTAPI

AI Generated News | <https://app.chaingpt.org/news>

2023

- Developed an AI-driven news platform for crypto, blockchain, and Web3, streamlining critical information delivery.
- Scraped news from top sources like Decrypt, CoinDesk, and CoinTelegraph and many more, enhanced with social signals from Twitter.
- Utilized embeddings to match news content across sources and timeframes, storing unique article embeddings in a vector database to avoid redundancy and ensure the publication of fresh content.
- Built the system to operate autonomously, incorporating a self-checking mechanism to maintain news accuracy and reliability.

Framework: Python, numpy, BeRT, chromadb, openai, selenium

Sentiment Analysis Web App for Political Feedback

August 2023

- Developed a FastAPI-based web app for sentiment analysis of political feedback, including aspect-based sentiment analysis.
- Leveraged advanced language models like Transformers and BERT for sentiment analysis.
- Identified positive and negative sentiments, preferences, and common themes in feedback.
- Associated feedback with specific political parties to provide contextual insights.
- Utilized large language models (LLM) to generate thoughtful conclusions based on feedback.

Framework: Pytorch, ChatGPT, BeRT, Transformers, Aspect-Based Sentiment Analysis, FastAPI

Multimodal title Classification

December 2022

- Developed a robust news headline classification system.
- Web scraping using Selenium for news titles and image URLs.
- Conducted extensive text preprocessing to enhance headline quality.
- Designed and implemented a sophisticated architecture for generating text embeddings and image feature extraction system to capture relevant features.
- Built a powerful machine learning model that combined text embeddings and image features for effective title classification.
- Integrated the system with a FastAPI endpoint for easy access.

Framework: Pytorch, Classification, Web Scraping, Selenium, FastAPI, Computer Vision, NLP.

AI Based Automatic Stationery Billing System

November 2022

- Application for automatic bill calculation of stationery items i.e. pencil, eraser, sharper, and scale.
- YoloV5 model is being used for object detection.
- On the backend prices of each object is set and the bill is calculated and displayed.

Framework: Pytorch, Yolo-v5, OpenCV

Skills

Programming	Python, SQL, JAVA, C/C++, R, LaTeX
Framework	PyTorch, Transformers, Deep Learning, Langchain, sklearn, Pandas, HuggingFace, FASTAPI, FLASKAPI, numpy
Cloud Computing	AWS, GCP, Azure, vast.ai, lambdalabs, AWS SageMaker
Databases	MySQL, chromadb, FAISS.
IDE	Anaconda, Spyder, Jupyter, IntelliJ IDEA, Eclipse, Visual Studio
Operating Systems	Ubuntu, Windows

Honors & Awards

Academic

2016 - 2020

NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES, PAKISTAN

- Two Gold medals, One Silver medal and One Bronze medal in the academic semester
- One Bronze medal in overall bachelors degree